

Impact of cryotherapy on airway stenosis: *In vivo*, *ex vivo*, and *in vitro* studies

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ABSTRACT

Cryotherapy is a promising technique for treating benign central airway stenosis (BCAS), but its underlying mechanisms remain unclear. This study investigated cryoablation effects on stenosis recurrence using *in vivo*, *ex vivo*, and *in vitro* models. Airway stenosis was induced in fourteen pigs, divided into balloon dilatation only (BD) and balloon dilatation with cryoablation (BD + C) groups ($n = 7$). Pigs were observed using bronchoscopy and computed tomography. Postmortem analyses included histopathology, Western blot, and quantitative RT-PCR (qPCR). *In vitro* studies using human lung fibroblasts (MRC-5 cells) assessed fibrosis-related markers and cellular responses using Western blot, qPCR, immunocytochemistry, and migration assays. *In vivo* observation revealed reduced stenosis progression in the BD + C group. *Ex vivo* analysis showed significantly lower histopathology scores for inflammation, epithelial denudation, and fibrosis, plus reduced fibrosis-related markers including collagen I and III, fibronectin, α -smooth muscle actin (α -SMA), and transforming growth factor- β 1 (TGF- β 1) in the BD + C group at protein and mRNA levels. *In vitro* studies using TGF- β 1-treated MRC-5 cells exposed to freezing showed reduced collagen I and α -SMA expression and impaired migration compared to controls. Cryoablation may inhibit stenosis progression by reducing TGF- β 1-regulated extracellular matrix deposition and fibrosis. These findings provide mechanistic insights into cryotherapy's therapeutic benefits for BCAS management.

1. Introduction

Benign central airway stenosis (BCAS) is a challenging condition characterized by narrowing of the trachea or main bronchi, potentially leading to significant morbidity and mortality (Ernst et al., 2004; Murgu et al., 2016). The etiology of BCAS is diverse and encompasses iatrogenic causes, including endotracheal intubation and tracheostomy; inflammatory conditions, such as granulomatosis with polyangiitis and sarcoidosis; infectious diseases, including tuberculosis; and post-procedural complications after airway stenting or surgical anastomosis (Gelbard et al., 2015; Murgu et al., 2016).

BCAS management typically involves surgical resection when feasible; however, endoscopic interventions are often necessary in

patients unsuitable for surgery (Ernst et al., 2004; Ost et al., 2015). These endoscopic approaches include balloon dilatation, electrocautery, argon plasma coagulation (APC), laser therapy, stenting, and cryotherapy (DiBardino et al., 2016; Flannery et al., 2018; Murgu et al., 2016). These treatments may cause airway injuries, leading to recurrent stenosis, particularly heat coagulation, such as electrocautery, APC, and laser therapy. These treatments can severely injure the airway mucosa and cartilage, potentially leading to additional scar tissue formation, airway restenosis, and treatment failure (Flannery et al., 2018). Thus, the challenge in treating BCAS lies not only in achieving initial patency but also in preventing restenosis.

Cryotherapy is a technique that uses “extremely cold temperatures” for diagnostic (e.g. tissue biopsy) or therapeutic purposes (e.g. foreign

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body removal, tumor removal, airway recanalization, or tumor ablation) (Colt, 2023; DiBardino et al., 2016; Sunna, 2013). It can be categorized into adhesion-based cryotherapy and tissue or cell destruction-based cryotherapy (Colt, 2023; DiBardino et al., 2016). Adhesion-based cryotherapy utilizes the property of the cryoprobe tip to adhere to the surrounding tissue during rapid cooling in a moist environment and is used for biopsy, tumor removal, airway recanalization, and the removal of foreign bodies, blood clots, and mucous plugs (Hetzel et al., 2004; Schumann et al., 2010). Tissue- or cell-destruction-based cryotherapy exploits cellular damage through intra- and extra-cellular crystallization and thrombus formation during freeze-thaw cycles, enabling tissue devitalization (Browning et al., 2013; Colt, 2023; Jung et al., 2016; Krinsky et al., 2010b). Tissue- or cell-destruction-based cryotherapy is called cryoablation and takes several days to take effect.

Cryoablation, a tissue- or cell-destruction-based cryotherapy method, is used to treat BCAS (Jung et al., 2016). Several studies have reported encouraging outcomes with cryoablation for BCAS, demonstrating relatively high remission rates (Browning et al., 2013; Colt, 2023; Finley et al., 2012; Krinsky et al., 2010a, 2010b). However, the mechanisms underlying the beneficial effects of cryotherapy in BCAS remain poorly understood. To address this gap, we designed a comprehensive study using *in vivo*, *ex vivo*, and *in vitro* models. The primary objective was to investigate whether cryoablation effectively prevents stenosis recurrence. This study provides insights into the potential clinical efficacy and cellular mechanisms of cryoablation in the treatment of airway stenosis.

2. Methods

2.1. Study design

The study was conducted in three parts: an *in vivo* study using a pig model, *ex vivo* analysis of pig tracheal samples, and *in vitro* studies using human lung fibroblasts (MRC-5 cells). The overall experimental flowchart is presented in [Supplementary Fig. S1](#). All animal experiments were approved by the Institutional Animal Care and Use Committee of Pusan National University Yangsan Hospital (approval no. P2023-008-A1C2). All experimental procedures were performed in accordance with the relevant guidelines and regulations for the care and use of laboratory animals. The study was conducted and reported in accordance with the ARRIVE guidelines (Percie du Sert et al., 2020). The *in vitro* experiments using MRC-5 cells were performed in accordance with the guidelines approved by the Institutional Review Board of Ulsan University Hospital (approval no. NON2024-010). Detailed methodological protocols for all procedures are described in the [Supplementary Materials](#).

2.2. Animal model and tracheal stenosis induction

Twelve-week-old female farm pigs (weight, 40–45 kg; tracheal diameter, 16–20 mm) were used due to the similarity of their tracheal size with humans (Judge et al., 2014). General anesthesia was administered via an intramuscular injection of 5 mg/kg alfaxan and maintained with 3 % isoflurane inhalation. Tracheal stenosis was induced using a validated model combining endotracheal cuff overpressure (200–400 mmHg for 40–60 min) and electrocautery applied at 8 circumferential spots using a coagulation suction tube (10390BN; Karl Storz, Germany) via a rigid bronchoscope (size 8.5, 10318BP; Karl Storz), as previously described by Kim et al. (2021). After inducing tracheal stenosis, the pigs were observed for one week to allow for stenosis development. Stenosis was considered successful when bronchoscopy or computed tomography (CT) confirmed ≥ 50 % reduction in airway cross-sectional area.

2.3. Study groups and intervention protocols

No formal sample size calculation was performed due to lack of prior large-animal data. This exploratory study with pilot-study characteristics was designed to assess feasibility and estimate effect magnitudes for future studies. A sample size of 7 animals per group was determined based on resource and ethical constraints.

Fourteen pigs with tracheal stenosis were divided into two intervention groups: balloon dilatation only (BD group, $n = 7$) and balloon dilatation with cryoablation (BD + C group, $n = 7$). Balloon dilatation was performed using a balloon catheter (CRE PRO wire guided, 12–13.5–15 mm, length 5.5 cm, catheter size 7.5 Fr; Boston Scientific, USA) inflated to a diameter of 15 mm for 1 min through a rigid bronchoscope. In the BD + C group, cryoablation was performed immediately after balloon dilatation using a flexible cryoprobe (length 900 mm, reusable $\varnothing 2.4$ mm or single use $\varnothing 1.7$ mm; ERBE Elektromedizin GmbH, Germany) applied at 4 circumferential positions (3, 6, 9, and 12 o'clock) with three 30-s freeze-thaw cycles at each position (Sunna, 2013). Follow-up examinations were performed weekly using flexible bronchoscopy and/or CT to assess tracheal diameter and stenosis progression. The degree of stenosis was quantified as the percentage reduction in tracheal lumen cross-sectional area compared to baseline measurements. During the observation period, pigs exhibiting dyspnea, pain, costal retraction, or significant tracheal stenosis were euthanized using intravenous potassium chloride injection. Pigs without signs of breathing difficulty or fatal tracheal stenosis were euthanized after 28–35 days using the same method.

2.4. Histopathological and molecular analyses

After euthanasia, tracheal specimens were harvested and processed for histopathological examination using hematoxylin and eosin (H&E) and Masson's trichrome staining. An experienced pathologist blinded to the treatment groups assessed inflammation, epithelial denudation, and fibrosis using a semi-quantitative scoring system (0–3 scale) as detailed in [Supplementary Table S1](#) (Feng Chen et al., 2022). The scoring scale ranged from 0 (absent) to 3 (severe), with higher scores indicating more severe pathological changes.

Western blot analysis and quantitative RT-PCR (qPCR) were performed on harvested tracheal tissue to evaluate the expression of key fibrosis-related markers, including collagen I, collagen III, fibronectin, α -smooth muscle actin (α -SMA), and transforming growth factor- $\beta 1$ (TGF- $\beta 1$) at both protein and mRNA levels (Wynn and Ramalingam, 2012).

2.5. *In vitro* studies with human lung fibroblasts (MRC-5 cells)

MRC-5 cells were cultured in α -minimum essential medium supplemented with 10 % fetal bovine serum (FBS) and 1 % penicillin-streptomycin at 37 °C in a 5 % CO₂ environment. To induce myofibroblast differentiation, cells were treated with 10 ng/mL recombinant human TGF- $\beta 1$ in media containing 2.5 % FBS for 5 days. To simulate cryoablation, cells were subjected to a freezing protocol involving exposure to precooled ethanol at -80 °C for 1 min, followed by thawing at 37 °C. Cell viability assessments confirmed that cryotherapy-treated cells maintained sufficient viability for subsequent analyses, with substantial recovery by 120 h post-treatment ([Supplementary Fig. S2](#)).

The effects of cryotherapy on fibrosis-related markers were assessed using Western blot analysis, qPCR, and immunocytochemistry for collagen I and α -SMA. For immunocytochemistry, cells were fixed with 4 % formaldehyde, permeabilized, and incubated with specific antibodies, followed by fluorescent imaging and quantitative analysis using confocal microscopy. Cell migration capacity was evaluated using a scratch assay, where cell monolayers were scratched and imaged at 0 and 24 h. The percentage of gap closure was quantified using ImageJ software to assess the migratory potential of cryotherapy-treated versus

untreated fibroblasts.

2.6. Statistical analysis

Data are presented as means $\pm$ standard errors (SEs) or medians with interquartile ranges (IQRs). Between-group comparisons were performed using the Mann-Whitney *U* test. Standardized effect sizes (Hedges' *g*) were calculated for key outcomes. Statistical significance was set at $p < 0.05$. Analyses were performed using SPSS version 24.0 (IBM Corp., Armonk, NY, USA).

3. Results

3.1. Effect of cryoablation on tracheal stenosis in vivo

Fourteen pigs were divided into two groups: BD ($n = 7$) and BD + C ($n = 7$). During stenosis induction, five pigs required a second induction procedure owing to insufficient initial stenosis ($<50\%$ reduction): two in the BD group (pigs 9 and 12) and three in the BD + C group (Pigs 3, 4, and 10) (Supplementary Table S2).

In the BD group, two pigs died suddenly on post-intervention day (PID) 0 and 5, and while the two required euthanasia due to severe respiratory distress on PID 7 and 12. In the BD + C group, two pigs died on PID 1 and PID 12, and one pig was euthanized owing to severe respiratory distress on PID 7 (Supplementary Table S2).

Post-intervention survival time showed a longer trend in the BD + C group than in the BD group (days, mean $\pm$ SE: 15.4 ± 4.6 vs. 17.9 ± 4.2 ; median [IQR]: $12.0 [5.0-28.0]$ vs. $21.0 [7.0-28.0]$, $p = 0.710$) (Table 1). Fig. 1 shows the progression of tracheal stenosis in representative pigs from each group. The BD + C group consistently showed larger tracheal diameters than the BD group from PID 0–28 (Fig. 2). At PID 21 and 28, the mean $\pm$ SE of the tracheal diameters in the BD + C group were 11.5 ± 1.0 mm and 12.7 ± 0.9 mm, respectively, compared to 8.3 ± 1.2 mm and 8.0 ± 1.5 mm in the BD group ($p = 0.114$ and 0.100), corresponding to large standardized effects (Hedges' *g* = 1.34 and 1.75, respectively). The BD + C group also showed lower percentages of stenosis throughout the study period. Pigs treated with cryoablation had less stenosis progression (final stenosis (%), mean $\pm$ SE: 81.4 ± 5.2 vs. 68.4 ± 7.1 ; median [IQR]: $81.0 [75.0-92.0]$ vs. $75.0 [53.0-84.0]$, $p = 0.209$). The standardized effect size for the final stenosis was moderate-to-large (Hedges' *g* = 0.74).

3.2. Histopathological changes in pig trachea

Histopathological examinations were performed on 11 pigs (BD group: 5; BD + C group: 6). Fig. 3 presents representative histological images and a qualitative analysis of the pathological features. The BD group exhibited severe neutrophilic inflammation, abundant granulation tissue formation, total epithelial denudation, and severe fibrosis. Conversely, the BD + C group exhibited mild neutrophilic inflammation, focal epithelial denudation, and mild fibrosis.

Semi-quantitative analysis revealed that the BD + C group had significantly lower inflammation scores than that of the BD group (mean $\pm$ SE [range]: $1.3 \pm 0.2 [1-2]$ vs. $2.0 \pm 0.3 [1-3]$), epithelial denudation (mean $\pm$ SE [range]: $1.2 \pm 0.4 [0-3]$ vs. $2.8 \pm 0.2 [2-3]$), and fibrosis (mean $\pm$ SE [range]: $1.2 \pm 0.2 [1-2]$ vs. $2.5 \pm 0.2 [2-3]$) (Fig. 4 and Supplementary Table S3). The cumulative histopathological scores further emphasized this difference, with the BD + C group having a significantly lower total score (mean $\pm$ SE [range]: $3.7 \pm 0.5 [2-5]$) compared to the BD group (mean $\pm$ SE [range]: $7.3 \pm 0.5 [6-9]$). Effect size analyses demonstrated large standardized differences between groups: inflammation (Hedges' *g* = 1.11), epithelial denudation (Hedges' *g* = 1.90), fibrosis (Hedges' *g* = 2.53), and cumulative histopathological score (Hedges' *g* = 2.81).

3.3. Expression of fibrosis-related markers in pig trachea

Western blot analysis and qPCR were performed on 6 pigs (BD group: 3, BD + C group: 3) to evaluate the expression of key fibrosis-related markers in tracheal samples from both groups.

Western blot analysis revealed a consistent trend towards a reduction in fibrosis-related protein expression in the BD + C group compared to the BD group (Fig. 5A). Densitometric analysis quantified these differences (Fig. 5B): Collagen I (mean $\pm$ SE: 3.1 ± 0.4 vs. 1.2 ± 0.1 , $p = 0.100$), collagen III (mean $\pm$ SE: 3.9 ± 0.4 vs. 0.7 ± 0.4 , $p = 0.100$), fibronectin (mean $\pm$ SE: 2.4 ± 0.2 vs. 1.0 ± 0.2 , $p = 0.100$), α -SMA (mean $\pm$ SE: 3.2 ± 0.3 vs. 1.3 ± 0.3 , $p = 0.100$), and TGF- β 1 expressions (mean $\pm$ SE: 3.8 ± 0.6 vs. 1.0 ± 0.4 , $p = 0.100$) were lower in the BD + C group than in the BD group. All protein levels were normalized to β -actin, which showed consistent expression across all samples. Standardized effect sizes for protein expression were large across all markers: collagen I (Hedges' *g* = 3.07), collagen III (Hedges' *g* = 3.69), fibronectin (Hedges' *g* = 2.68), α -SMA (Hedges' *g* = 2.45), and TGF- β 1 (Hedges' *g* = 2.78).

The qPCR analysis showed trends consistent with the results of the protein expression analysis at the transcriptional level (Fig. 5C).

Table 1

Comparison of tracheal diameter, stenosis rate, and survival between balloon dilatation only and balloon dilatation with cryoablation groups.

		Balloon dilatation only			Balloon dilatation with cryoablation		<i>p</i> -value*
		Mean $\pm$ SE	Median(IQR)		Mean $\pm$ SE	Median(IQR)	
Tracheal inner diameter, mm	Day 0 ($n = 7$)	16.6 ± 0.5	17.0(15.0–18.0)	Day 0 ($n = 7$)	17.4 ± 0.8	18.0(15.0–19.0)	0.383
	PID 0 ($n = 7$)	8.7 ± 0.9	8.0(7.0–12.0)	PID 0 ($n = 7$)	9.1 ± 0.8	10.0(7.0–11.0)	0.902
	PID 7 ($n = 5$)	7.3 ± 1.4	8.5(5.0–10.0)	PID 7 ($n = 6$)	9.8 ± 1.3	9.5(7.5–11.0)	0.329
	PID 14 ($n = 3$)	8.7 ± 1.3	10.0(6.0–10.0)	PID 14 ($n = 4$)	12.0 ± 1.2	12.0(10.5–13.5)	0.229
	PID 21 ($n = 3$)	8.3 ± 1.2	9.0(6.0–10.0)	PID 21 ($n = 4$)	11.5 ± 1.0	11.5(10.0–13.0)	0.114
	PID 28 ($n = 3$)	8.0 ± 1.5	9.0(5.0–10.0)	PID 28 ($n = 3$)	12.7 ± 0.9	13.0(11.0–14.0)	0.100
Stenosis, %	Day 0 ($n = 7$)	0 (Baseline)		Day 0 ($n = 7$)	0 (Baseline)		
	PID 0 ($n = 7$)	71.6 ± 4.9	78.0(56.0–80.0)	PID 0 ($n = 7$)	72.1 ± 4.0	70.0(67.0–82.0)	1.000
	PID 7 ($n = 5$)	77.8 ± 7.5	75.0(69.0–92.0)	PID 7 ($n = 6$)	67.8 ± 9.0	73.5(63.0–84.0)	0.537
	PID 14 ($n = 3$)	71.0 ± 9.3	69.0(56.0–88.0)	PID 14 ($n = 4$)	54.3 ± 10.0	57.5(41.0–67.5)	0.400
	PID 21 ($n = 3$)	73.7 ± 7.3	69.0(64.0–88.0)	PID 21 ($n = 4$)	58.5 ± 8.2	61.5(48.0–69.0)	0.229
	PID 28 ($n = 3$)	74.0 ± 10.1	75.0(56.0–91.0)	PID 28 ($n = 3$)	50.7 ± 7.9	53.0(36.0–63.0)	0.200
Final stenosis, %	($n = 7$)	81.4 ± 5.2	81.0(75.0–92.0)	($n = 7$)	68.4 ± 7.1	75.0(53.0–84.0)	0.209
Survival after intervention, days	($n = 7$)	15.4 ± 4.6	12.0(5.0–28.0)	($n = 7$)	17.9 ± 4.2	21.0(7.0–28.0)	0.710

* *p*-value were analyzed using the Mann–Whitney *U* test.
SE, standard error; PID, post-intervention day.

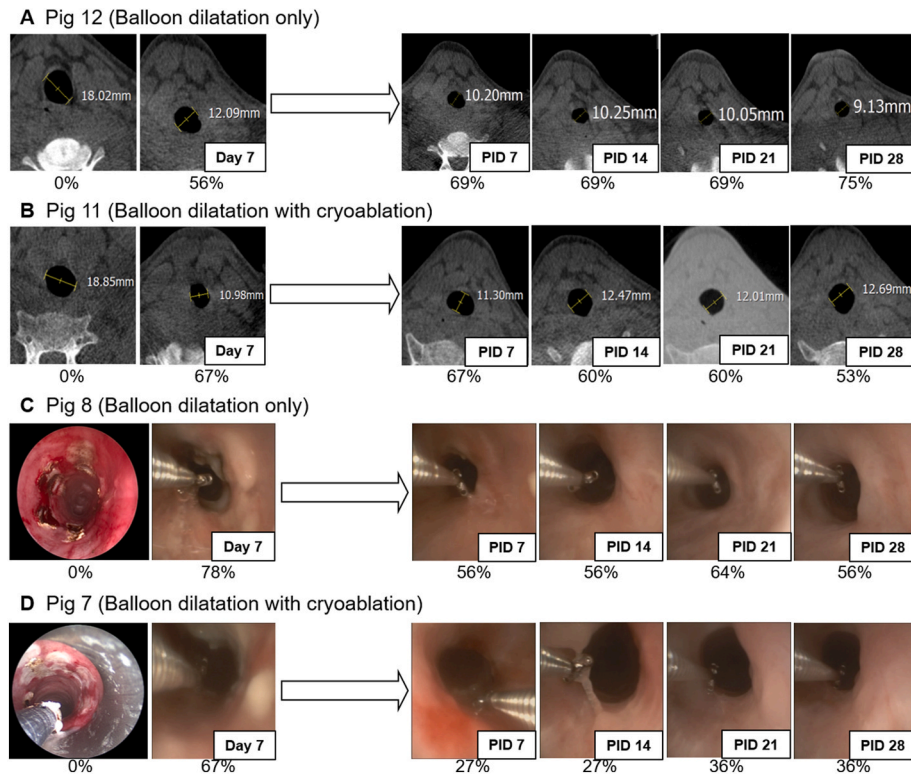


Fig. 1. Representative CT scans and bronchoscopic images
A; CT images of Pig 12 treated with balloon dilatation only.
B; CT images of Pig 11 treated with balloon dilatation and cryoablation.
C; Bronchoscopic images of Pig 8 treated with balloon dilatation only.
D; Bronchoscopic images of Pig 7 treated with balloon dilatation and cryoablation.
 Abbreviations: CT, computed tomography.

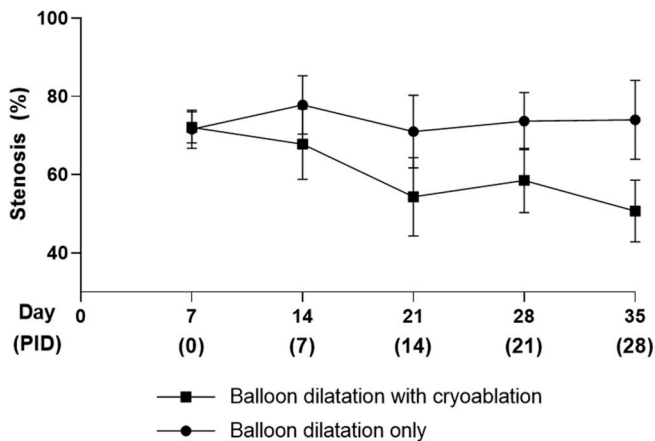


Fig. 2. Effects of cryoablation on tracheal stenosis progression
 Trends in the tracheal stenosis rate (%) observed over 4 weeks following intervention. The solid circles (●) represent the balloon dilatation only group, while the solid squares (■) represent the balloon dilatation with cryoablation group. The intervention was performed on day 7 (PID: 0). Data are represented as means $\pm$ standard errors.
 Abbreviation: PID, post-intervention day.

Collagen I, collagen III, fibronectin, and α -SMA mRNA levels showed a 3.4-fold (mean $\pm$ SE: 5.1 ± 1.2 vs. 1.5 ± 0.5 , $p = 0.100$), 2.8-fold (mean $\pm$ SE: 5.8 ± 1.2 vs. 2.1 ± 0.5 , $p = 0.100$), 3.5-fold (mean $\pm$ SE: 2.5 ± 0.6 vs. 0.7 ± 0.2 , $p = 0.100$), and 3.2-fold (mean $\pm$ SE: 2.0 ± 0.5 vs. 0.6 ± 0.1 , $p = 0.100$) reduction, respectively, in the BD + C group relative to the BD group. Furthermore, TGF- β 1 mRNA expression showed the most

pronounced decrease, with a 5.1-fold reduction in the BD + C group compared to the BD group (mean $\pm$ SE: 1.7 ± 0.4 vs. 0.3 ± 0.1 , $p = 0.100$). Standardized effect sizes for mRNA expression were large across all markers: collagen I (Hedges' $g = 1.55$), collagen III (Hedges' $g = 1.60$), fibronectin (Hedges' $g = 1.64$), α -SMA (Hedges' $g = 1.70$), and TGF- β 1 (Hedges' $g = 2.10$).

3.4. Effect of cryotherapy on TGF- β 1-treated MRC-5 cells

To further investigate the cellular mechanisms of cryotherapy, we conducted *in vitro* experiments using TGF- β 1-treated MRC-5 cells.

Western blot analysis revealed that cryotherapy reduced the expression of both collagen I and α -SMA compared to untreated cells (Fig. 6A). Densitometric analysis quantified these differences (Fig. 6B): Collagen I expression was lower in cryotherapy-treated cells than in untreated cells (cryo-vs. cryo + [mean $\pm$ SE]: 0.9 ± 0.1 vs. 0.4 ± 0.1 , $p = 0.029$). Similarly, α -SMA levels were reduced in cryotherapy-treated cells relative to untreated cells (cryo-vs. cryo + [mean $\pm$ SE]: 0.9 ± 0.0 vs. 0.4 ± 0.1 , $p = 0.029$). These findings were corroborated at the transcriptional level with qPCR (Fig. 6C). Cryotherapy treatment resulted in the decreased mRNA expression of collagen I and α -SMA in treated cells compared to untreated cells (cryo-vs. cryo + [mean $\pm$ SE]: 0.9 ± 0.2 vs. 0.2 ± 0.0 , $p = 0.008$ and 1.0 ± 0.1 vs. 0.6 ± 0.0 , $p = 0.008$, respectively).

Immunocytochemistry analysis further supported these results, demonstrating reduced expression and altered cellular distribution of collagen I (red) and α -SMA (green) in cryotherapy-treated cells compared to untreated cells (Fig. 6D). The quantitative analysis of fluorescence intensity revealed a significant decrease in both collagen I and α -SMA expression in treated cells (Fig. 6E). The mean fluorescence

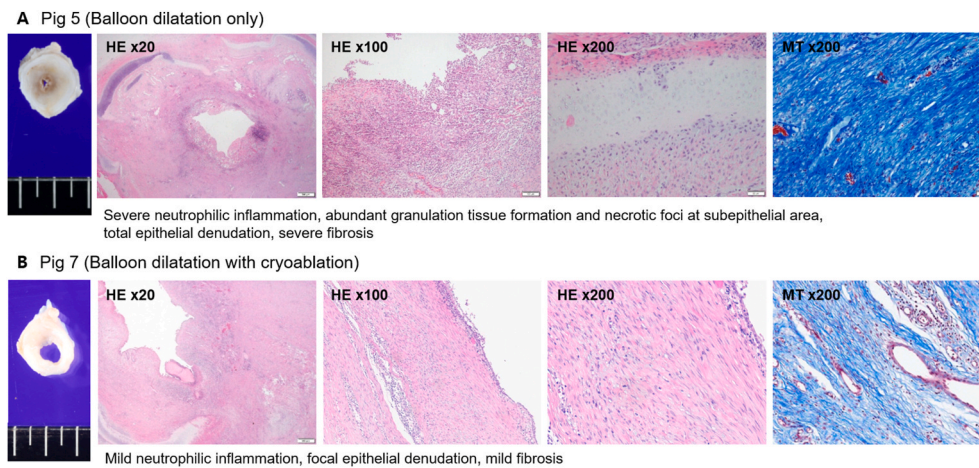


Fig. 3. Representative histopathologic images

A; Trachea of Pig 5 (balloon dilatation only) exhibiting severe neutrophilic inflammation, severe epithelial denudation, and severe fibrosis. H&E staining at 200 × showing extensive neutrophilic infiltration and total epithelial denudation. MT staining at 200 × highlighting significant fibrotic tissue.

B; Trachea of Pig 7 (balloon dilatation with cryoablation) exhibiting mild neutrophilic inflammation, focal epithelial denudation, and mild fibrosis. H&E staining at 200 × showing mild neutrophilic infiltration and focal epithelial denudation. MT staining at 200 × showing mild fibrotic tissue.

Abbreviations: H&E, hematoxylin and eosin stain; MT, Masson's trichrome stain.

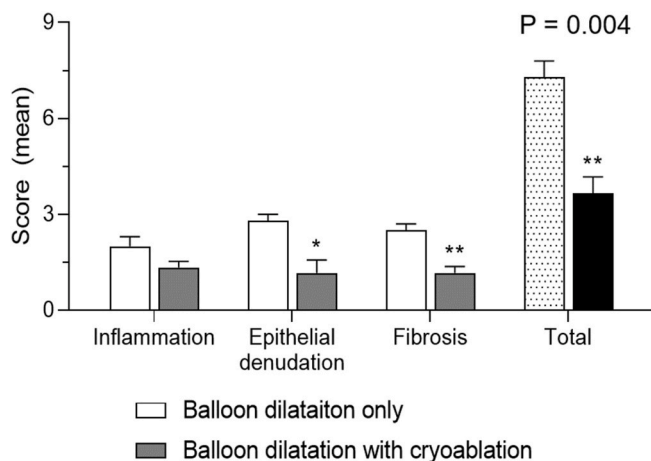


Fig. 4. Comparison of histopathological scores between balloon dilatation only (white bars) and balloon dilatation with cryoablation (gray bars) groups. Data are presented as mean ± standard error. * $p < 0.05$ and ** $p < 0.01$ indicate statistically significant differences.

intensity for collagen I and α -SMA was lower in cryotherapy-treated cells than in untreated cells (cryo-vs. cryo + [mean ± SE]: 123.0 ± 12.2 vs. 73.5 ± 5.2 , $p = 0.017$ and 587.0 ± 72.7 vs. 381.2 ± 27.0 , $p = 0.032$, respectively).

To assess the functional impact of cryotherapy on surviving cells, a scratch assay on TGF- β 1-treated MRC-5 cells was performed. After 24 h, cryotherapy-treated cells exhibited a significantly lower migration rate than that of untreated cells (cryo-vs. cryo + [mean ± SE]: 79.3 ± 2.9 vs. 28.8 ± 6.5 , $p = 0.002$) (Fig. 7). These results demonstrate that cryotherapy affects both the expression of fibrosis-related markers and the migratory capacity of TGF- β 1-treated MRC-5 cells.

4. Discussion

Our study investigated the potential of cryoablation as an adjunct therapy for managing BCAS, using complementary *in vivo*, *ex vivo*, and *in vitro* approaches to establish both clinical efficacy and underlying molecular mechanisms.

Previous clinical studies have reported varying outcomes of

cryoablation in benign airway stenosis. A recent systematic review of 67 post intubation or tracheostomy tracheal stenosis cases demonstrated clinical improvement in 72 % of patients with minimal complications after cryotherapy (Hosna et al., 2023), while another study involving 26 patients with benign tracheal stenosis reported an 84 % reduction in stenosis after spray cryotherapy (Bhara et al., 2016). However, these studies lacked mechanistic explanations for the observed improvements. Our study addresses this knowledge gap by comprehensively analyzing cryoablation's effects through multiple experimental approaches.

Our *in vivo* pig model showed that the combination of balloon dilatation and cryoablation resulted in consistently lower percentages of tracheal stenosis than balloon dilatation alone. Although these differences were not statistically significant, likely due to the small sample size, they are consistent with previous clinical studies that have reported positive outcomes of cryotherapy in airway stenosis (Finley et al., 2012; Krinsky et al., 2010b).

The *ex vivo* histopathological analysis revealed significantly reduced inflammation, epithelial denudation, and fibrosis in the BD + C group compared to the BD group. These findings suggest that cryoablation may influence the underlying pathophysiology of tracheal stenosis. Inflammation plays a crucial role in fibrotic tissue remodeling through multiple signaling pathways (Bao et al., 2020; Wang et al., 2025), suggesting that cryoablation's anti-inflammatory effects may interrupt the progression from acute injury to chronic fibrotic stenosis. The observed reductions in these key pathological features could potentially contribute to improved outcomes and help prevent restenosis, a common challenge in BCAS management (Hillel et al., 2014).

Our molecular studies provided critical insights into the mechanisms underlying these clinical effects. The consistent reduction in fibrosis-related markers (collagen I, collagen III, fibronectin, and α -SMA) at both protein and mRNA levels in the BD + C group suggests that cryoablation modulates the fibrotic process through multiple pathways. Notably, TGF- β 1 expression was markedly reduced in the BD + C group, suggesting this key mediator of fibrosis, implicated in the pathogenesis of tracheal stenosis (Karagiannidis et al., 2006; Zhang et al., 2023), is an important target of cryoablation therapy. The interplay between inflammatory mediators and fibrotic pathways (Bao et al., 2020) supports the notion that cryoablation's anti-inflammatory effects may contribute to its antifibrotic properties.

The therapeutic effects of cryoablation likely result from controlled freezing, which selectively modifies tissue responses through direct cellular injury and vascular effects. This leads to altered protein

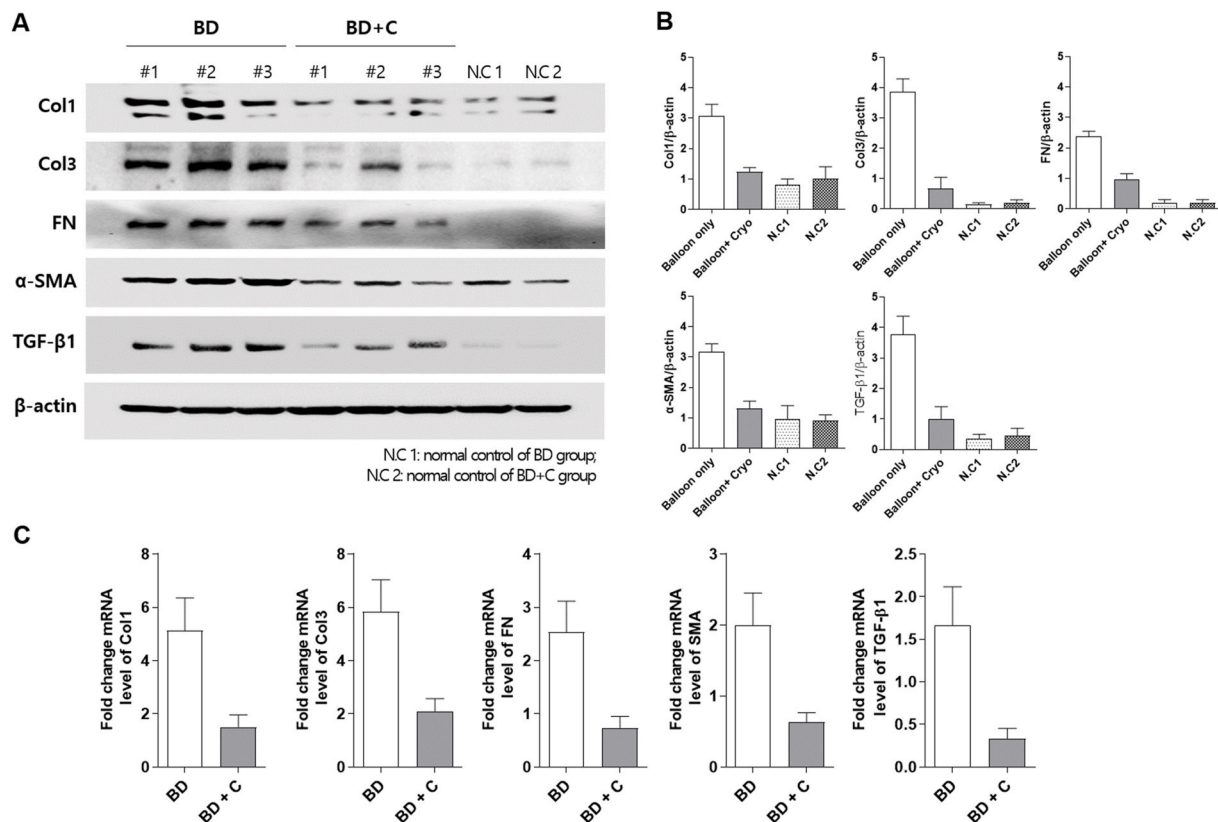


Fig. 5. Comparison of fibrosis-related markers in pig trachea

A; Western blot images showing the expression of Col1, Col3, FN, α-SMA, TGF-β1, and β-actin in tracheal tissue samples.

B; Quantitative analysis of protein expression levels normalized to β-actin.

C; mRNA expression levels of fibrosis-related genes (Col1, Col3, FN, α-SMA, TGF-β1) in pig trachea.

Data are represented as means ± standard errors. P-values were calculated compared to the balloon dilatation only group.

Abbreviations: Col1, collagen I; Col3, collagen III; FN, fibronectin; α-SMA, α-smooth muscle actin; TGF-β1, transforming growth factor-β1.

expression patterns and cytokine networks, particularly within the TGF-β1 pathway (Baust et al., 2014; Erinjeri and Clark, 2010), providing a mechanistic framework for understanding how cryotherapy achieves antifibrotic effects in airway stenosis.

Our *in vitro* studies using MRC-5 cells provided further molecular evidence for the anti-fibrotic effect of cryoablation. Cryotherapy significantly reduced both collagen I and α-SMA expression at protein and mRNA levels. The reduction in α-SMA expression following cryotherapy indicates decreased myofibroblast differentiation, a critical process in stenosis development (Hinz et al., 2007). This finding is particularly relevant as myofibroblasts are the predominant matrix-producing cells in fibrotic conditions (Darby et al., 2014; Kendall and Feghali-Bostwick, 2014).

The functional impact of cryotherapy was further demonstrated in the migration assays, where treated fibroblasts showed significantly reduced motility. Since fibroblast migration is crucial for fibrosis development (Darby et al., 2014; Kendall and Feghali-Bostwick, 2014), its inhibition by cryotherapy, combined with reduced fibrotic marker expression, suggests comprehensive anti-fibrotic effects at the cellular level.

The antifibrotic mechanisms of cryoablation observed in this study may have implications beyond tracheal stenosis treatment, particularly for dermatological applications such as hypertrophic scars and keloids (O'Boyle et al., 2017). Previous studies have shown that keloid fibroblasts are more susceptible to cryotherapy-induced apoptosis than normal fibroblasts (Dalkowski et al., 2003). These insights could guide improvements in cryotherapy protocols for managing various fibrotic conditions (Gupta and Sharma, 2011; Kim, 2021).

While previous studies have investigated spray cryotherapy as a

single treatment modality for airway stenosis (Bhora et al., 2016), our study examined a novel combination of balloon dilatation and cryoablation. This approach provides complementary therapeutic effects: cryotherapy targets the fibrotic response through TGF-β1 signaling modulation, while mechanical dilatation offers immediate improvement in airway patency.

Cryoablation offers advantages over traditional treatments for BCAS, such as balloon dilatation alone or thermal ablation techniques (electrocautery, APC, or laser), which often provide only temporary relief and are associated with high restenosis rates (Ernst et al., 2004; Murgu et al., 2016). Unlike thermal ablation, cryoablation preserves the extracellular matrix structure, potentially facilitating healing and reducing restenosis risk (Erinjeri and Clark, 2010; Gage and Baust, 1998).

This study has several limitations. First, no formal sample size calculation was performed, as this exploratory study with pilot characteristics lacked prior large-animal data for reliable estimation. The sample size was determined by resource and ethical constraints inherent to large-animal research. While statistical power was limited for some outcomes, effect size analyses demonstrated moderate-to-large effects, indicating biologically meaningful differences. Second, the absence of a cryotherapy-only group limits isolation of cryotherapy's independent effects. However, this design reflects clinical practice where balloon dilatation precedes cryotherapy for immediate airway patency, as cryoablation effects develop gradually. Given the delayed therapeutic action and the early mortality observed in our model, a cryotherapy-only group would likely have experienced substantially higher mortality without immediate mechanical relief. Despite these limitations, the convergence of *in vivo* trends with significant *ex vivo* and *in vitro* findings

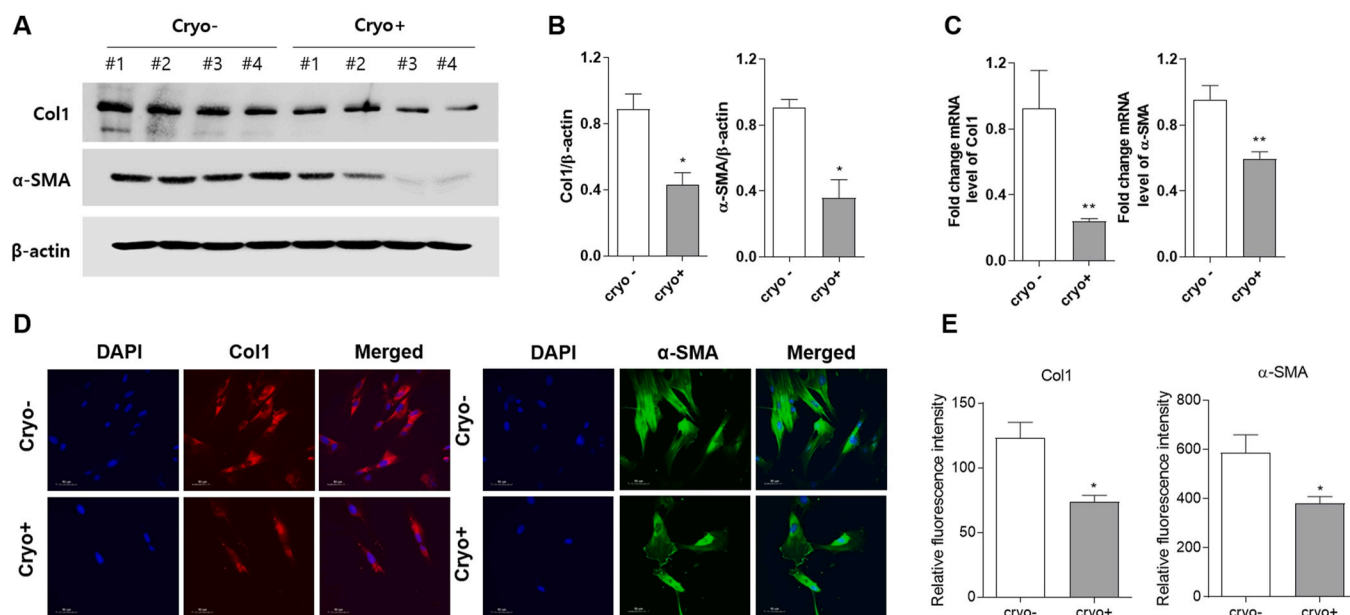


Fig. 6. Effects of cryotherapy on col1 and α-SMA expression in MRC-5 cells treated with TGF-β1
A; Western blot images showing the expression of Col1, α-SMA, and β-actin.
B; Quantitative analysis of protein expression levels normalized to β-actin.
C; mRNA expression levels of fibrosis-related genes (Col1, α-SMA).
D; Confocal immunocytochemistry fluorescence images showing Col1 (red) and α-SMA (green) expression levels. Scale bar: 50 μm.
E; Quantitative analysis of Col1 and α-SMA fluorescence intensity.
 Data represent the means ± standard errors. * $p < 0.05$, ** $p < 0.01$ indicate statistically significant differences.
 Abbreviations: Col1, collagen I; α-SMA, α-smooth muscle actin; TGF-β1, transforming growth factor-β1.

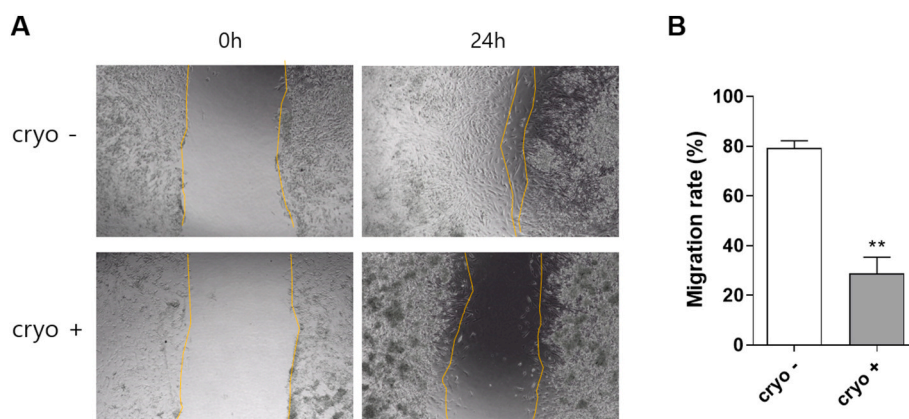


Fig. 7. Effects of cryotherapy on migration of MRC-5 cells treated with TGF-β1
A; Representative images of scratch assays showing MRC-5 cells immediately after (0 h) and 24 h post-scratch procedure, comparing cryotherapy-treated and untreated conditions (scale bar: 200 μm).
B; Migration distance was quantified by the gap closure percentage after 24 h.
 Data represent means ± standard errors. * $p < 0.05$, ** $p < 0.01$ indicate statistically significant differences.
 Abbreviations: Col1, collagen I; α-SMA, α-smooth muscle actin; TGF-β1, transforming growth factor-β1.

provides multi-modal evidence for cryoablation's antifibrotic mechanisms, establishing proof-of-concept for future confirmatory studies.

5. Conclusions

This study provides preliminary evidence that cryoablation combined with balloon dilatation may effectively treat BCAS by modulating TGF-β1 signaling and suppressing fibrotic processes. The observed trends in stenosis reduction *in vivo*, decreased fibrosis-related marker expression in *ex vivo* samples and impaired fibroblast migration *in vitro*, suggest that cryoablation offers a biologically targeted approach to

preventing restenosis. These findings provide a foundation for further investigations into cryoablation's efficacy in BCAS management.

CRedit authorship contribution statement

Ganghee Chae: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Ok-Joo Sul:** Writing – original draft, Methodology, Investigation, Formal analysis. **Chul Park:** Writing – review & editing. **Jin Hyoung Kim:** Writing – review & editing, Investigation, Formal analysis. **Seung Won Ra:** Writing – review & editing, Methodology. **Kwang Won Seo:** Writing

– review & editing. **Yangjin Jegal**: Writing – review & editing, Methodology. **Jong Joon Ahn**: Writing – review & editing. **Hee Jeong Cha**: Writing – review & editing, Validation, Methodology, Formal analysis. **Taehoon Lee**: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare no competing interests.

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None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jtherbio.2026.104399>.

Data availability

Data will be made available on request.

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